Question Paper

Exam Date & Time: 18-Nov-2019 (10:00 AM - 01:00 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

I Semester of M. Sc. (AOC) Examination
November 2019
Date: 18/11/2019 Day: Monday Time: 10.00 am to 1.00 pm

PHYSICAL CHEMISTRY-I [OC743]

Marks: 70 **Duration: 180 mins.**

MCQ

Answer all the questions. Section Duration: 30 mins

Choose the correct answer

1) What is the efficiency of engine operated between 120 ^oC to 30 ^oC? (1)

23 % 25 % 30 % 40 %

2) What is the value of ΔS when one mole of ideal gas expanded at 300 K from 10 L to 100 L ? (1)

20 J 19 J 18 J 17 J

3) What is the size of the particle for colloids? (1)

1 to 10 A 10 to 2000 A Above 2000 A Below 10 A

4) Which of the following condition represent an isochoric process? (1)

ΔV > 0 ΔP = 0 ΔV = 0 ΔP = 1

5) The magnitude of adsorption is increases with increase in ______. (1)

Temperature Heat Volume Pressure

6) The correct dimension for gas constant R is ___. (1)

lit atm. $mol^{-1}K^{-1}$ lit atm^{-1} mol. K lit^{-1} atm^{-1} mol. K lit atm. mol. K

7) If the plot of x / a(a - x) vs. time is straight line with positive slope then value of intercept is __. (1)

One Two Three Zero

8) If the cell constant of conductivity cell is 0.1 cm^{-1} and resistance is 2 ohm, then what is the value of specific conductance? (1)

0.5 5.5 0.05 50

9) What is the general equation for unit of rate constant k ? (1)

$(Mol./lit)^{n-1}$ S^{-1} $(Mol./lit)^{1-n}$ S^{-1} $(Mol./lit)^{n}$ S^{-1} $(Mol./lit)^{n}$ S

10) The rate of adsorption is directly proportional to ___. (1)

θ P (2 - θ) P (3 - θ)T (1 - θ) P

11) For the reaction of H_2SO_4 to SO_2 how many electrons are transfer? (1)

6 2 4 0

12) What is the equation for relation between partition function and energy? (1)

- $E = -N\, d\ln q/d\beta$
- $E = -N\, d\ln \beta/dq$
- $E = -\beta\, d\ln q/dN$
- $E = N\, d\ln q/d\beta$

13) What is the unit of rate constant for the second order reaction? (1)

- $Mol^{1}.lit^{-1}.s^{-1}$
- $Mol^{-1}.lit^{1}s^{-1}$
- $Mol^{1}.lit.\ s^{-1}$
- $Mol^{1}.lit^{-1}$

14) What is the value of weight configuration when 20 objects are distributes in the arrangement (0, 1, 5, 0, 8, 3, 2, 1) ? (1)

- 3.2×10^{10}
- 5.2×10^{10}
- 6.2×10^{10}
- 4.2×10^{10}

15) What is the dimension for de Broglie wave length is? (1)

- L
- M
- T
- S

16) For enzyme catalyzed reaction the order of reaction is, if Km = [S] ___. (1)

- First
- Zero
- Second
- Half

17) Which order reaction is not completed 100 %? (1)

- Second
- First
- Third
- Zero

18) The number of reactant molecules that participate in elementary process is called ___. (1)

- Order of reaction
- Rate constant
- Molecularity
- Rate of reaction

19) For second order reaction the value of rate constant is 0.09 $M^{-1}sec^{-1}$ and initial concentration is 0.1 M then what the value of half-life time is ? (1)

- 121 min
- 131 min
- 111 min
- 141 min

20) The substance which is absorbed on the surface of another substance is called ____. (1)

- Sorption
- Adsorbate
- Adsorbent
- Absorption

Section I

Answer all the questions.

21) Derive the equation for second order reaction with concentration of both reactant are different and discuss its characteristics. (5)

22) Find out the entropy change when 10 gm. of tin heated from 293 K to 573 K. The melting point of tin is 505 K, heat of fusion is 14cal/g, specific heat of solid tin and liquid tin are 0.055 and 0.064 respectively. (5)

[OR]
23) Discuss the vapour pressure curve for non-ideal solution which exhibits positive and negative deviation from ideal behavior. (5)

24) Derive the relation $q_R = 8\pi^2\, I\, KT/h^2$ (5)

25) For diatomic molecules derive the relation for vibrational partition function (5)

[OR]
26) Obtain the relation for entropy, weight configuration and partition function. (5)

Section II

Answer all the questions.

27) What is the consecutive reaction? Derive the equation for consecutive reaction (6)

28)	What is order reaction? Derive the relation for integration rate law for n^{th} order reaction.	(4)
[OR] 29)	Discuss the relation between activation energy and rate constant k and discuss its characteristics	(4)
30)	For the reaction in standard cell the value of emf = 1.0185 V at 25 ^{o}C and dE^{o}/dT is 5×10^{-5} V/K. Calculate the value of ΔG, ΔH and ΔS	(5)
31)	What is concentration cell? Discuss the electrode concentration cell with suitable example.	(5)
[OR] 32)	Derive the relation $E = E^{o} + (RT/nF) \ln K$.	(5)
33)	Write the differences between physisorption and chemisorption.	(6)
34)	Discuss the method for testing of Freundlich adsorption isotherm.	(4)
[OR] 35)	Derive the equation $P/y = 1/a + (b/a) P$ and discuss its characteristics.	(4)

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